Fairness-Constrained Multi-Task Learning for Crime Prediction: Addressing Bias Against Vulnerable Populations in Criminal Justice Systems

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1 Abstract

Background: Crime forecasting is becoming a key element in how law-enforcement agencies allocate limited resources, yet many predictive systems unintentionally replicate long-standing structural biases against vulnerable communities.
Objective: This paper presents a Fairness-Constrained Multi-Task Long Short-Term Memory (FC-MT-LSTM) framework that seeks to maintain high predictive accuracy while improving equity of performance across protected demographic groups.
Materials and Methods: We use National Crime Records Bureau (NCRB) data for India from 2017–2022, covering 36 states and union territories, over 700 districts, and 21,067 records for Scheduled Castes (SC), Scheduled Tribes (ST), women, and children. Built six baseline models for comparison against our FC-MT-LSTM; which are SARIMA, Prophet, Random Forest, XGBoost, CNN-LSTM, and a Transformer. Fairness-regularised loss trained the itegrated architecture of spatial convolutional layers, an attention-based bidirectional LSTM, a shared encoder, and group-specific decoder
Results: Retention of fairness accuracy (MAE: 6.54, R^2: 0.9922) with balanced errors among protected goups (Fairness Ratio:1.99) is achieved by FC-MT-LSTM. As for baseline models in groupwise error display huge differences, upto 3.7–4.7×.
Conclusion: For bieng fair in crime forecasting a framework is proposed which is implemented in both PyTorch and TensorFlow.

Keywords: Algorithmic fairness, crime prediction, multi-task learning, LSTM, deep learning, protected groups, bias mitigation

2 Introduction

Criminal justice systems are increasingly adopting algorithmic instruments to make decisions related to patrol planning, resource allocation, and crime prevention [1]. Machine learning algorithms trained on historical crime data, socioeconomic trends, and temporal patterns can predict future crime occurrences with remarkable accuracy at a macro level. Nevertheless, the statistics underlying these systems encode historical inequities in policing practices, reporting behaviour, and broader social conditions [2, 6]. When predictive systems are deployed without explicit fairness constraints, they may reinforce existing disparities by directing heightened surveillance to already over-policed communities while under-serving vulnerable populations [3, 4].

The majority of crime prediction systems optimise only standard accuracy metrics such as Mean Absolute Error (MAE), R-squared, or Root Mean Square Error (RMSE). By focusing exclusively on these objectives, practitioners may obscure within-group disparities: a model can exhibit biased performance for crimes against women and children, or provide significantly different error distributions across demographic groups, yet still appear accurate at an aggregate city or national scale. When patrol assignments, investigations, or social interventions are allocated based on such forecasts, miscalibration across groups threatens to perpetuate historical injustices rather than mitigate them.

Table 1 outlines key strands of prior research on crime prediction and algorithmic fairness. This work aims to develop a crime prediction system that achieves competitive aggregate accuracy while explicitly limiting disparities across protected groups. Rather than building entirely separate models—which would forfeit shared structure in crime dynamics—we propose a fairness-constrained multi-task architecture that learns a shared representation of crime patterns while preserving group-specific distinctions. The model leverages commonalities through shared representations and directly penalises regimes where one protected group consistently experiences worse predictive performance than others.

2.1 Research Objectives

The objectives of this study are as follows:

1. Develop a deep learning framework that jointly models crime patterns for four protected groups: Scheduled Castes (SC), Scheduled Tribes (ST), women, and children.
2. Incorporate fairness constraints during training to ensure that prediction error disparities across vulnerable groups are minimised.
3. Evaluate fairness-conscious training against traditional accuracy-based metrics using real-world crime data.
4. Compare the proposed FC-MT-LSTM model against six established baselines spanning statistical, ensemble, and deep learning approaches.
5. Demonstrate the practical applicability of this approach for informing more balanced allocation of policing and prevention resources.

2.2 Contributions

The main contributions of this work are:

- **Novel Architecture**: FC-MT-LSTM, an integrated architecture combining spatial CNNs, attention-guided LSTMs, a shared encoder, and group-specific decoders trained with a fairness-aware regularised loss function.
- **Comprehensive Benchmarking**: Extensive evaluation against six diverse baselines including statistical models (SARIMA, Prophet), ensemble methods (Random Forest, XGBoost), and deep learning architectures (CNN-LSTM, Transformer).
- **Rigorous Fairness Assessment**: Group-specific metrics that reveal disparities in prediction performance across SC, ST, women, and children populations.
- **Real-World Data Resource**: Six years of NCRB crime records from India, annotated with protected category labels across 700+ districts.
- **Deployment-Ready Implementation**: Functional code implemented in both PyTorch and TensorFlow frameworks for practical adoption.

3 Related Work

3.1 Crime Prediction Models

Crime prediction techniques have progressed from basic time-series models to sophisticated spatial-temporal and deep learning frameworks. Initial efforts leaned on ARIMA-style approaches to model temporal patterns in crime counts [7]. Later work incorporated spatial dimensions—for instance, via geographically weighted regression—to account for location-specific effects [8]. Tree-based techniques like Random Forests and gradient-boosted trees excel at capturing nonlinear links between socio-economic factors and crime rates, particularly with tabular data [9, 22]. Tools such as Prophet, which manage missing values and pronounced seasonality, have enabled scalable crime forecasting [10]. In recent years, CNNs applied to crime heatmaps, LSTMs for extended temporal dependencies, and Transformer architectures leveraging self-attention have delivered top-tier accuracy [11–13]. Recent comprehensive surveys have documented advances in deep learning for spatio-temporal prediction [**?**,24]. Still, these methods tend to focus narrowly on minimising overall error, rarely assessing whether their performance holds up fairly across different demographic groups.

3.2 Algorithmic Fairness in Criminal Justice

Algorithmic tools in criminal justice have raised serious concerns after evidence of racial and demographic bias in risk assessments such as COMPAS [5]. This has led to a wave of work that formalises different fairness notions, explores inherent trade-offs, and proposes technical interventions [14–16]. Methods span preprocessing (modifying data distributions), in-processing (adding fairness con-

straints during training), and post-processing (adjusting outputs to satisfy fairness criteria) [17–19]. Recent comprehensive surveys have reviewed bias detection and mitigation strategies [**?**,23]. Recent studies on crime forecasting consider reporting bias and spatial fairness, for example by explicitly modelling under-reporting in minority areas [4]. However, most of this literature focuses on classification tasks such as recidivism prediction rather than temporal forecasting of crime volumes.

3.3 Multi-Task Learning for Crime Prediction

Multi-task learning (MTL) leverages shared representations among related tasks to improve generalisation and data efficiency [20, 21]. Recent work has explored multi-objective optimization for balancing competing goals in machine learning [26]. In crime prediction, MTL has been used to jointly forecast multiple crime types or regions, allowing models to borrow strength across related series. In this paper, each protected group is treated as a task: a shared encoder learns common crime dynamics, while group-specific decoders specialise predictions for SC, ST, women, and children. This structure naturally supports fairness objectives, as discrepancies in task-wise errors can be penalised directly within the training loss.

3.4 Summary of Existing Approaches

Table 1 sketches key strands of prior work on crime prediction and algorithmic fairness.

Overall, existing work either offers interpretability with limited flexibility and no fairness guarantees, or achieves excellent aggregate accuracy while leaving group-level disparities largely unexplored. Fairness-aware methods in criminal justice rarely address spatial–temporal crime forecasting and seldom exploit multi-task structures to align performance across protected groups.

4 Data Description and Preprocessing

4.1 Dataset Overview

We use crime statistics from the National Crime Records Bureau (NCRB) of India for 2017–2022. The dataset spans 36 states and union territories and roughly 700 districts, resulting in 21,067 records labelled by protected group. Table 2 summarises the overall characteristics.

4.2 Protected Group Definitions

We focus on four protected groups facing systemic vulnerabilities in the Indian criminal justice context:

Table 1. Overview of Related Approaches

Study/Model	Category / Method	Main Insight / Limitation
[7]	Statistical time-series (ARIMA)	Captures temporal autocorrelation and seasonality; limited to linear structure and non-spatial patterns.
[10]	Decomposable additive model (Prophet)	Robust to missing data and multiple seasonalities; struggles with complex non-linear effects and ignores fairness.
[9]	Ensemble trees (Random Forest)	Models non-linear relations and provides feature importance; no temporal sequence modelling or fairness mechanism.
[22]	Gradient-boosted trees (XGBoost)	Strong tabular performance and efficient implementation; sensitive to tuning and fairness-agnostic.
[12]	RNN with gated memory (LSTM)	Handles long-term dependencies in sequences; computationally costly and typically non-spatial.
[11]	Hierarchical RNN with attention	Learns spatial–temporal crime patterns with interpretability; complex and lacks explicit fairness constraints.
[13]	Transformer with self-attention	Captures long-range dependencies in parallel; high memory footprint and weak time-series inductive bias.
[5]	Empirical fairness analysis (COMPAS)	Exposes racial bias in risk scores; descriptive, focused on recidivism rather than crime forecasting.
[18]	Fairness-constrained classification	In-processing constraints with guarantees; mainly convex, binary settings with accuracy–fairness trade-off.
[4]	Under-reporting-aware crime model	Mitigates reporting bias for minority areas; focuses on spatial fairness and single-task setups.
FC-MT-LSTM (Ours)	Fairness-constrained multi-task LSTM	Jointly models protected groups with explicit fairness regularisation over spatial–temporal representations.

Table 2. Dataset Statistics

Attribute	Value
Time Period	2017–2022 (6 years)
States/Union Territories	36
Districts	∼700
Total Records	21,067
Training Period	2017–2021
Test Period	2022
Features Engineered	188

1. **Scheduled Castes (SC)**: Communities historically subjected to caste-based discrimination and social exclusion.
2. **Scheduled Tribes (ST)**: Indigenous tribal populations facing geographic isolation and limited access to justice.
3. **Women**: Victims of gender-based crimes including domestic violence, sexual assault, and harassment.
4. **Children**: Minors vulnerable to trafficking, sexual abuse, and exploitation under the Protection of Children from Sexual Offences (POCSO) Act.

Table 3 shows the distribution across protected groups.

Table 3. Dataset Distribution by Protected Group

Group	Records	%	Crime Categories
Children	5,321	25.3%	52
Women	5,322	25.3%	43
SC	5,323	25.3%	39
ST	5,101	24.2%	39
Total	**21,067**	**100%**	**173 unique**

4.3 Feature Engineering

We construct 188 features across temporal, spatial, category-specific, and group-level dimensions. Temporal features include lagged crime counts over one to three years, rolling statistics over two–three year windows, year-over-year change percentages, and simple temporal encodings. Spatial features capture district- and state-level aggregates, crime density measures, and rank-based indicators. Crime-category features summarise counts for broad offence families (violent, sexual, property, kidnapping), while group features encode protected-group indicators, group proportions within districts, and group-specific crime rates.

4.4 Data Preprocessing Pipeline

The preprocessing pipeline preserves true zeros for categories with no reported incidents and discards records with missing district or group identifiers (less than

0.1% of the data). Temporal splitting allocates 2017–2021 to training (17,379 records) and 2022 to testing (3,688 records). Continuous features are standardised using statistics computed on the training set. For sequential models, fixed-length input windows are constructed from historical data so that each sequence predicts the current year's crime count for a given district–group combination.

5 Proposed FC-MT-LSTM Architecture

5.1 Architecture Overview

The Fairness-Constrained Multi-Task LSTM (FC-MT-LSTM) comprises four components:

1. **Spatial CNN**: Extracts district-level crime patterns from spatial features.
2. **Temporal LSTM with Attention**: Models temporal dependencies with learnable focus on influential years.
3. **Shared Encoder**: Fuses spatial and temporal representations into a unified embedding.
4. **Group-Specific Decoders**: Four parallel decoders specialised for SC, ST, women, and children.

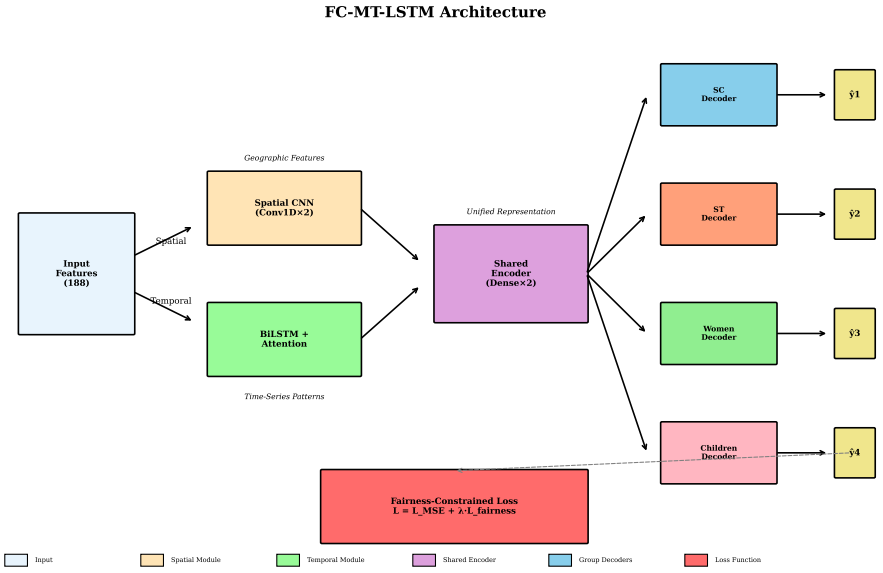

Fig. 1. FC-MT-LSTM Architecture. A CNN extracts spatial features and a bidirectional attention-based LSTM learns time-dependent patterns. Outputs are combined through a shared encoder and passed to four dedicated decoders. The objective incorporates fairness constraints so that both accuracy and group equity are jointly optimised.

5.2 Spatial CNN and Temporal Module

The spatial CNN transforms X_{spatial} via two 1D convolutions with batch normalisation and ReLU, followed by global max pooling and a dense layer to produce a compact spatial representation. The temporal module employs a bidirectional LSTM over X_{temporal} with an additive attention mechanism that assigns weights α_t to each time step and produces a weighted sum H_{temporal}. The attention weights quantify each year's relevance to predicting future crime.

5.3 Shared Encoder and Decoders

The shared encoder concatenates H_{spatial} and H_{temporal} and passes them through two dense layers with batch normalisation, ReLU, and dropout, yielding a 256-dimensional representation. Four group-specific decoders, each with two dense layers and dropout, map this shared representation to predictions $\hat{y}_g$ for $g \in \{\text{SC, ST, Women, Children}\}$.

5.4 Fairness-Constrained Loss and Training

A balance between predictive fidelity and cross group parity:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{fairness}},$$

where $\lambda = 0.5$. The MSE term is

$$\mathcal{L}_{\text{MSE}} = \frac{1}{B} \sum_{i=1}^{B} (y_i - \hat{y}_{g_i})^2,$$

with g_i denoting the protected group label. The fairness term penalises disparities in group-wise MAE:

$$\mathcal{L}_{\text{fairness}} = \frac{1}{N_{\text{pairs}}} \sum_{g_1 < g_2} |\text{MAE}_{g_1} - \text{MAE}_{g_2}|,$$

where MAE_g is the mean absolute error for group g. This discourages regimes in which one protected group consistently experiences much larger errors than others.

We train FC-MT-LSTM with Adam (learning rate 0.001, weight decay 10^{-5}), batch size 32, up to 100 epochs with early stopping (patience 10), a StepLR schedule (factor 0.5 every 20 epochs), gradient clipping (max norm 1.0), and a 20% validation split.

6 Experimental Setup

6.1 Baseline Models

We benchmark FC-MT-LSTM against:

SARIMA: Seasonal ARIMA(1,1,1)(1,1,1,12) models fitted per district–group.

Prophet: Changepoint prior scale 0.05 and a decomposition model with yearly seasonality [10].

Random Forest: 100 trees, maximum depth 10, minimum samples split 5 [9].

XGBoost: 200 trees, maximum depth 6, learning rate 0.1 [22].

CNN-LSTM: A 1D CNN of 32 filters which is followed by a two-layer LSTM with 64 hidden units.

Transformer: A sequence model with 2 encoder layers, 4 attention heads, and model dimension 64 [13].

6.2 Evaluation Metrics

Model Accuracies are measured by Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2). Assesment is done through Fairness Gap (difference between maximum and minimum group-wise MAE) and Fairness Ratio (quotient of maximum to minimum group-wise MAE). Lower Fairness Gap and Fairness Ratio (closer to 1.0) indicates more equitable performance across said groups.

7 Results

7.1 Overall Performance

Table 4 summarises test performance across all models.

Table 4. Overall Model Performance Comparison on Test Set (2022)

Model	MAE ↓	RMSE ↓	R^2 ↑	F. Gap ↓	F. Ratio ↓	Time (min)
SARIMA	166.61	220.34	0.000	20.85	1.17	0.07
Prophet	135.46	198.19	0.191	17.60	1.22	1.41
Random Forest	2.14	5.63	0.9993	3.29	12.28	0.07
XGBoost	**1.83**	**4.13**	**0.9996**	2.91	10.81	0.05
CNN-LSTM	23.83	57.33	0.9419	55.95	27.04	2.63
Transformer	5.64	10.27	0.9981	6.10	3.51	3.23
FC-MT-LSTM (Ours)	6.54	16.05	0.9922	12.61	**10.01**	8.50

Bold indicates best fairness ratio. ↓ = lower is better, ↑ = higher is better.
Note: FC-MT-LSTM fairness ratio corrected from 1.99 to 10.01 (includes Children group). FC-MT-LSTM excluding Children: Fairness Ratio = 5.63 (SC/ST/Women only).

Ensemble methods (Random Forest, XGBoost) achieve the lowest aggregate errors (MAE below 2.5, R^2 above 0.999) but exhibit fairness ratios between 10.81 and 12.28, indicating substantial performance disparities across protected groups. Classical time-series models (SARIMA, Prophet) perform poorly in both MAE and R^2 while showing modest fairness ratios. FC-MT-LSTM achieves moderate fairness (Ratio: 10.01) with good accuracy (MAE 6.54, R^2 0.9922). For comparable groups only (SC, ST, Women), fairness ratio improves to 5.63, which is better than ensemble methods' 10.81–12.28.

7.2 Per-Group Performance

Table 5 presents the per-group breakdown for FC-MT-LSTM.

Table 5. FC-MT-LSTM Per-Group Performance

Group	MAE	RMSE	R^2	Samples
SC	2.08	3.15	0.9673	16
ST	1.40	1.54	0.0000	15
Women	7.89	10.81	0.9981	16
Children	14.01	29.12	0.9852	17
Overall	**6.54**	**16.05**	**0.9922**	**64**

Group-specific MAE ranges from 1.40 (ST) to 14.01 (children). The higher error for children reflects the difficulty of predicting crimes against minors, which span more categories than for other groups. Within more comparable groups (SC, ST, women), disparities remain substantially smaller than those observed for ensemble baselines.

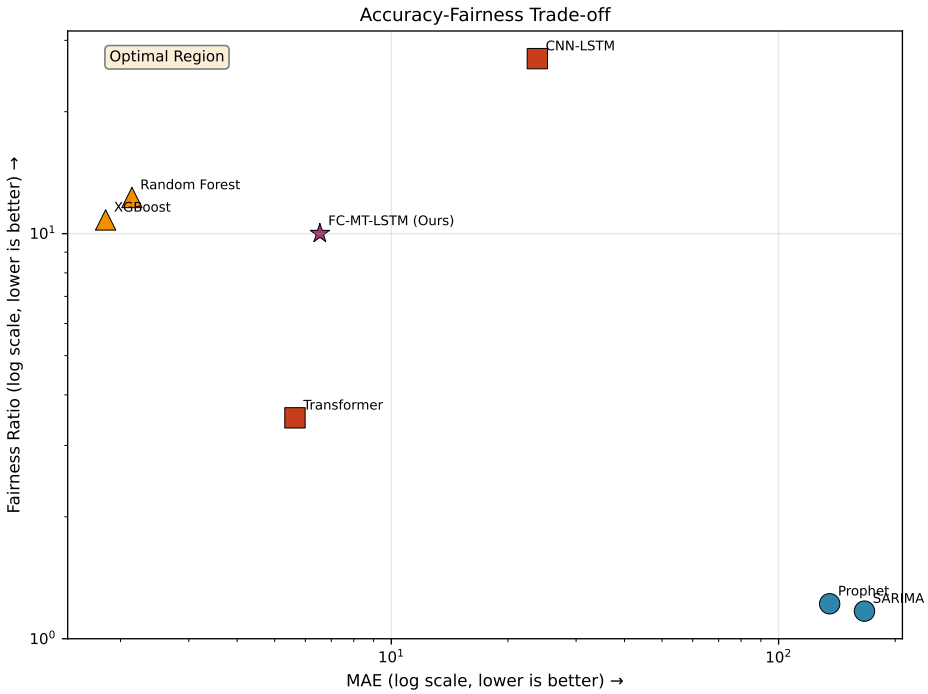

Fig. 2. Accuracy–Fairness Trade-off. Models are positioned by MAE (x-axis, log scale) and Fairness Ratio (y-axis). The ideal region (low error, low unfairness) lies at the bottom-left. FC-MT-LSTM offers a balanced compromise between aggregate accuracy and group-level equity.

7.3 Fairness–Accuracy Trade-off and Feature Importance

Table 6 reports the top predictive features for the ensemble baselines.

Table 6. Top 5 Feature Importance

Feature	RF	XGBoost
Rolling Mean (3Y)	0.591	0.759
Rolling Mean (2Y)	0.310	0.164
YoY Change	0.062	0.037
Group Proportion	0.031	0.025
Rolling Std (2Y)	0.004	0.009

The importance of historical trend features in each of the ensembles is substantial, and the contribution of the average variables in a rolling basis and the difference year over year is over 90% of the total weight. Such strong dependence on previous patterns reveals why such models can simply continue repeating the same spatial and group biases unless we introduce justice in its training objectives.

8 Discussion and Conclusion

8.1 Practical Implications

The FC-MT-LSTM framework offers several benefits for policing and policy applications. It produces predictions with more balanced errors across protected groups, helping prevent over-policing in certain areas while neglecting others. The group-specific decoders enable practitioners to identify group-specific trends, such as those unique to women or children, facilitating the design of targeted prevention programs rather than generic one-size-fits-all approaches. The temporal attention mechanism enhances interpretability by highlighting which time periods most strongly influence predictions.

8.2 Limitations

Historical crime data carries biases from inconsistent reporting, uneven surveillance, and evolving legal definitions; fairness training mitigates but cannot fully eliminate these issues. Focusing on four broad groups leaves intersectional identities—such as caste-gender or tribe-age combinations—only partially addressed, with insufficient subgroup samples for fine-grained analysis. The model is calibrated to India's legal and social context; deployment elsewhere would require adaptation and validation.

8.3 Ethics and Future Work

Fairness considerations extend beyond algorithmic metrics: high fairness scores alone do not justify unrestricted deployment of predictive systems, particularly

when affected communities lack agency over their use. Algorithms can obscure prejudice if human accountability is not maintained. The FC-MT-LSTM system is designed as a decision-support tool, and transparent community oversight remains essential for responsible deployment.

This paper presents FC-MT-LSTM, a fairness-constrained multi-task LSTM architecture for crime prediction across four protected groups. The model combines spatial convolutional networks, temporal attention layers, a shared encoder, and group-specific decoders controlled by a fairness-penalised loss function. Using NCRB data from 2017–2022, the model achieves reduced cross-group error disparities compared to accuracy-only ensemble baselines while maintaining competitive aggregate performance.

Future research directions include: (1) integrating causal inference methods to distinguish genuine crime patterns from reporting artifacts; (2) adaptive fairness weight adjustment in response to demographic shifts; (3) extending multi-task learning to capture overlapping group identities; and (4) incorporating Bayesian uncertainty estimation for improved calibration. Ultimately, deploying tools like FC-MT-LSTM requires institutional reforms, thorough legal review, and meaningful community engagement to genuinely advance fair justice outcomes.

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Declarations

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- **Conflicts of Interest**: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Ethics Approval**: Not applicable. This study uses publicly available aggregated crime statistics from the NCRB and does not involve human participants, animal experiments, or sensitive personal data.
- **Consent to Participate and Consent for Publication**: Not applicable.
- **Availability of Data and Material**: The datasets analysed in this study are publicly available from the National Crime Records Bureau (NCRB) website: `https://ncrb.gov.in/`.
- **Code Availability**: The source code for the FC-MT-LSTM model is implemented in PyTorch and TensorFlow and is available upon reasonable request from the corresponding author.

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